



A PROJECT REPORT ON

GYM MANAGEMENT SYSTEM

(Gym Core Pro)

Submitted in partial fulfilment of the requirements for the award of
BACHELOR OF COMPUTER APPLICATIONS / COMPUTER SCIENCE

SUBMITTED BY:

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CERTIFICATE

This is to certify that the project report entitled "**Gym Management System (Gym Core Pro)**" is a bona fide work carried out by **Umesh** (Roll No.: [Roll / Enrollment Number]) in partial fulfilment of the requirements for the award of degree/diploma during the academic session 2026.

The work presented in this report has been carried out under my direct supervision and guidance at **ZillionSoftech**. To the best of my knowledge, the matter embodied in this project report has not been submitted elsewhere for the award of any other degree or diploma.

I wish him every success in all his future academic and professional endeavors.

Date: _____

Place: _____

Mr. Kamal Kumar
Project Guide & Technical Mentor
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DECLARATION

I, **Umesh**, hereby declare that the project report entitled "**Gym Management System (Gym Core Pro)**" submitted by me in partial fulfilment of the requirements for the award of the degree is an authentic record of original work performed by me under the guidance of **Mr. Kamal Kumar** at **ZillionSoftech**.

The information, code architecture, database designs, and analysis presented in this document are true and original to the best of my knowledge. This project has not been submitted previously, either in part or full, to any other institute or university for the award of any degree, diploma, or certificate.

Umesh

Student / Developer

Date: _____

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to **ZillionSoftech** for providing me with the practical platform, resources, and technical support required to design and build the "**Gym Management System**" project.

I am profoundly thankful to my guide, **Mr. Kamal Kumar**, for his continuous guidance, insightful feedback, and constant encouragement throughout the development lifecycle. His expert guidance was instrumental in clarifying complex aspects of software architecture, Model-View-Controller (MVC) design patterns, and relational database management.

I also extend my sincere appreciation to the entire faculty, laboratory instructors, and my peers for providing a collaborative and encouraging environment during this work.

Umesh

ABSTRACT

The **Gym Management System (Gym Core Pro)** is a comprehensive desktop software application engineered in Python designed to automate core business operations for fitness centers, gymnasiums, and health clubs. It efficiently manages member enrollments, subscription plans, renewal lifecycles, and financial transactions through a secure centralized database.

By replacing fragmented manual registers and static spreadsheet logs, the system eliminates administrative overhead, mitigates financial leakage, and provides clear visibility into active memberships. Key components include a credential-authenticated login portal, an executive dashboard with key operational metrics, a full member CRUD interface, subscription expiry tracking, and transaction history logging.

Built using Python 3, Tkinter GUI, and MySQL Database with an enterprise Model-View-Controller (MVC) and Data Access Object (DAO) architecture, the solution delivers high performance, data validation, and modular maintainability.

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CHAPTER 1: INTRODUCTION

1.1 System Overview

Gym Management Systems govern how modern health clubs register members, configure flexible membership plans, track plan validity periods, and maintain billing records. Historically, small and medium fitness centers relied heavily on physical paper ledgers or standalone spreadsheets. As customer counts expand, these manual methods result in undetected expirations, uncollected dues, and tedious audit procedures.

The **Gym Management System** addresses these shortcomings by delivering a desktop application built upon a robust relational database engine, ensuring real-time data integrity and immediate retrieval.

1.2 Need for Automation

- **Centralization:** Consolidates member profiles, plans, and receipts into one accessible database.
- **Revenue Protection:** Automatically flags expired plans and unpaid balances.
- **Accountability:** Tracks payment timestamps and administrative actions.
- **Security:** Enforces login authentication to protect confidential customer data.

1.3 Scope of the Project

The system provides a complete administrative suite including member lifecycle management, membership plan setup, automated duration calculation, fee logging, dynamic search, and metric reporting.

CHAPTER 2: PROBLEM STATEMENT

Manual paper ledgers and unstructured spreadsheet solutions lead to serious operational challenges:

- Duplicate profile entries and inconsistent contact information.
- Members accessing facilities long after their plans have expired due to lack of real-time alerts.
- Difficulty auditing revenue logs and missing payment records.
- Data vulnerability and lack of administrative access control.

CHAPTER 3: LITERATURE REVIEW

Feature	Paper Ledgers	Spreadsheets	Cloud SaaS	Proposed Gym Core
Data Structure	Unstructured	Row/Column	Relational Cloud	Relational MySQL
Search Speed	Manual / Slow	Basic Filter	Fast Online	Instant Dynamic Search
Expiry Alert	None	Manual Color	Automated Web	Automated Engine
Operating Cost	Low	Low	High Recurring	Zero Licensing (Open-Source)

CHAPTER 4: REQUIREMENT ANALYSIS

4.1 Functional Requirements

1. **Authentication:** Secure login verification for system users.
2. **Member CRUD:** Complete creation, modification, viewing, and deletion of member records.
3. **Plan Management:** Definition of customized membership durations and rates.
4. **Payment Logging:** Recording financial transactions tied to specific member IDs.
5. **Dashboard Analytics:** Displaying key summary stats (Total Members, Monthly Revenue, Active Subscriptions).

CHAPTER 5: FEASIBILITY STUDY

Technical Feasibility: Python 3 and MySQL are well-supported industry standards with mature GUI libraries (Tkinter).

Economic Feasibility: Built entirely using free, open-source development tools (VS Code, Python, MySQL), resulting in zero software costs.

Operational Feasibility: Clean, minimalist interface designed for rapid adoption with minimal staff training.

CHAPTER 6: SYSTEM DESIGN

The system is architected using the **Model-View-Controller (MVC)** and **Data Access Object (DAO)** design patterns to isolate user interface components from core business logic and SQL persistence layers.

CHAPTER 7: DATABASE DESIGN

7.1 Table Structure: members

Column Name	Data Type	Constraints	Description
member_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique Member ID
full_name	VARCHAR(100)	NOT NULL	Full name of member
phone	VARCHAR(20)	NOT NULL	Validated contact number
email	VARCHAR(100)	DEFAULT NULL	Email address
join_date	DATE	NOT NULL	Registration date

CHAPTER 8: TECHNOLOGY STACK

- **Programming Language:** Python 3.10+
- **GUI Framework:** Tkinter / ttk
- **Database Engine:** MySQL 8.0 Server (InnoDB)
- **Database Connector:** mysql-connector-python
- **Development Environment:** Visual Studio Code

CHAPTER 9: PROJECT MODULES

1. **Auth Module:** Controls admin/staff login access.
2. **Member Module:** Handles customer registration, profile edits, and deletion.
3. **Plan Module:** Sets duration (Monthly, Quarterly, Annual) and pricing tiers.
4. **Payment Module:** Tracks transaction logs, payment modes, and outstanding balances.
5. **Dashboard Module:** Visual overview of operational metrics.

CHAPTER 10: IMPORTANT CODE SNIPPETS

10.1 Database Connection Configuration

```
import mysql.connector
from config import DB_CONFIG

class DatabaseConnection:
    @staticmethod
    def get_connection():
        return mysql.connector.connect(
            host=DB_CONFIG['host'],
            user=DB_CONFIG['user'],
            password=DB_CONFIG['password'],
            database=DB_CONFIG['database']
        )
```

10.2 Parameterized SQL Query Execution

```
from database.connection import DatabaseConnection

class MemberDAO:
    @staticmethod
    def add_member(full_name, phone, email, join_date):
        conn = DatabaseConnection.get_connection()
        cursor = conn.cursor()
        query = "INSERT INTO members (full_name, phone, email, join_date) VALUES (%s, %s, %s, %s)"
        cursor.execute(query, (full_name, phone, email, join_date))
        conn.commit()
        cursor.close()
        conn.close()
```

CHAPTER 11: TESTING

Test Case ID	Description	Expected Result	Pass/Fail
TC-01	User Authentication	Grants access on valid credentials	PASS
TC-02	Phone Validation	Rejects non-numeric phone entries	PASS
TC-03	Member Registration	Saves new record to MySQL table	PASS
TC-04	Payment Logging	Links payment record to Member ID	PASS

CHAPTER 12: RESULTS

The developed system successfully delivers a clean, desktop management environment enabling administrators to rapidly register members, track plan expiration dates, and monitor daily revenue with zero data corruption.

CHAPTER 13: ADVANTAGES

- Normalized MySQL database prevents data duplication.
- Parameterized SQL execution eliminates SQL Injection security vulnerabilities.
- Separation of concerns via MVC pattern ensures code maintainability.

CHAPTER 14: LIMITATIONS

- Local desktop deployment limits remote access without complex network setup.
- Does not include direct credit card processing payment gateway integration.

CHAPTER 15: FUTURE SCOPE

- Integration of SMS/WhatsApp APIs for automated plan expiration reminders.
- Biometric attendance integration (fingerprint / facial recognition).
- Web-based customer portal built with Flask/FastAPI and React.

CHAPTER 16: CONCLUSION

The **Gym Management System (Gym Core Pro)** achieves its foundational objective of digitizing and streamlining gym operations. Built under the mentorship of **ZillionSoftech**, the software offers a complete, secure, and robust platform suitable for commercial gym administration.

REFERENCES

1. Python Software Foundation, "Python 3 Documentation", <https://docs.python.org/3/>
2. Oracle Corporation, "MySQL 8.0 Reference Manual", <https://dev.mysql.com/doc/>
3. Elmasri, R., & Navathe, S. B., "Fundamentals of Database Systems", Pearson.
4. Sommerville, I., "Software Engineering", 10th Edition, Pearson.

APPENDIX

A.1 Project Directory Structure

```
gym_managment_system_umesh-main/
├── config.py
├── main.py
├── controllers/
│   ├── login_controller.py
│   └── member_controller.py
├── dao/
│   ├── member_dao.py
│   ├── payment_dao.py
│   └── user_dao.py
├── database/
│   ├── connection.py
│   └── gym_db.sql
├── models/
│   ├── member.py
│   └── user.py
├── utils/
│   ├── constants.py
│   ├── helper.py
│   └── validation.py
└── views/
    ├── dashboard.py
    ├── home_view.py
    ├── login_view.py
    ├── memberview_view.py
    ├── payment_view.py
    ├── plan_view.py
    ├── subscription_view.py
    └── user_view.py
```